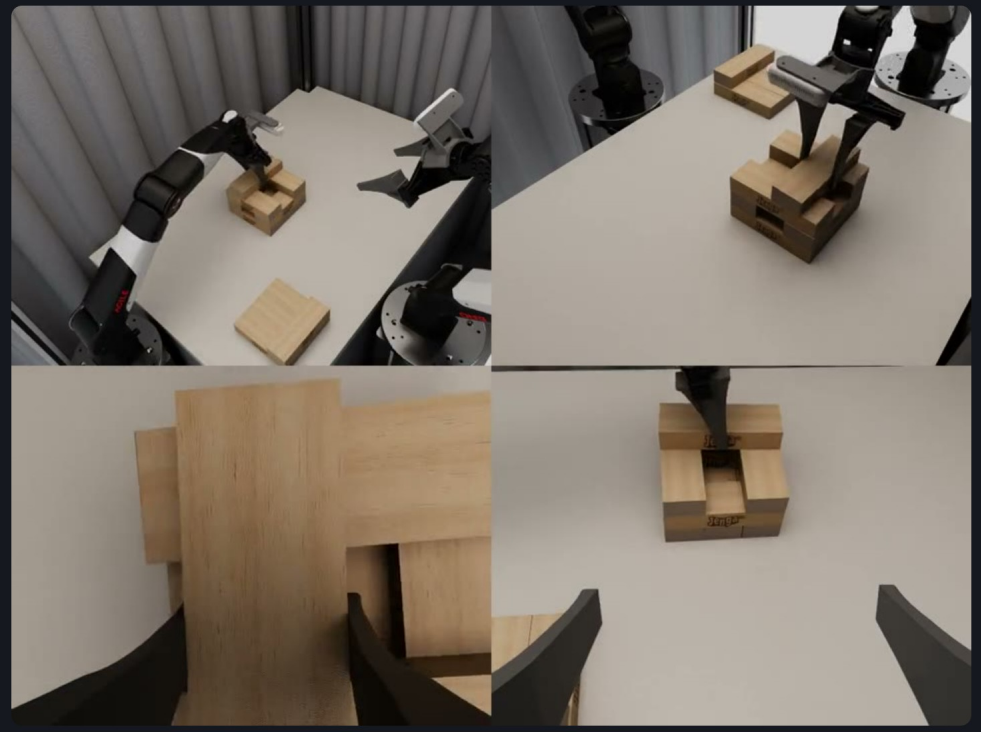
TOWER CHALLENGE 2026 · TEAM SAKAI

Agentic Task Decomposition for Skill-Specialist Robot Policies

Think slowly at design and training, act fast at run time.

Yifeng (Sakai) Liu



Official eval · RAX-AM v6d · T1-3 · 4×

OVERVIEW

Our path in three steps

STEP 1 · π0.5

End-to-end policy

π0.5, fully fine-tuned on the official demos. One vision-language-action model: images and joints in, joint commands out.

89.4 on Tasks 1–3, official
Stuck on Tasks 4–5

STEP 2 · RAX

Agent-supervised program

Coding agents split each task into skills and write, with us, a program that retargets the demos to the blocks it perceives.

frequently time out,
90 of 100 official episodes

STEP 3 · RAX-AM

Agent program + small models

The agents' program makes the training data. Small learned MLPs, 0.45–17 M parameters, pick and execute the skills.

91 of 100 official episodes
Best single submission 91.93

Agents think slowly at design time; small models act fast at run time.

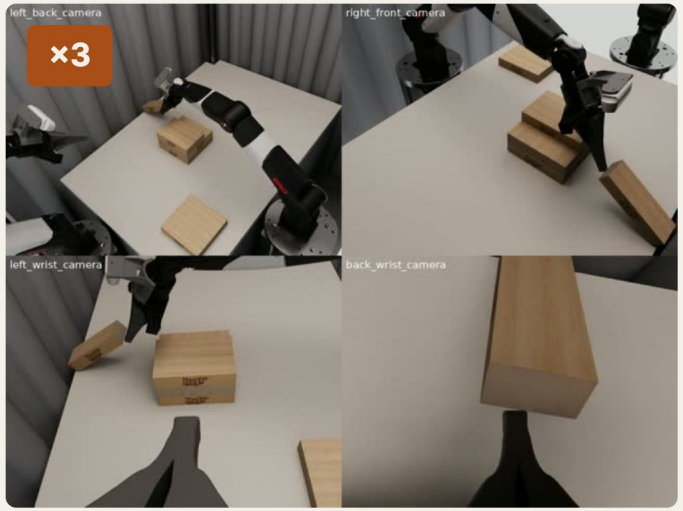
End to end: strong on Tasks 1–3, stuck on 4–5

89.4

π0.5 full fine-tune, Tasks 1–3 average score

≈38 h
fine-tuning, 4 stages

2× H20
mostly · ≈130 GPU-hours



Task 4 · push-down



Task 5 · insertion, about 1 mm per side

A necessary fix: the official static cameras were rotated 180°.

Tasks 4–5 count three times scores in this competition. In our replica, π0.5 never completed a Task 5 episode, and Task 4 plateaued at around 60% successful rate

02 · DESIGN TIME

Coding agents as System 2

Robot Agent Exploration for Action Models

OFFICIAL SCORE PER VARIANT · TASKS 4–5 ×3

$$s = 60 \cdot LHPR + 30 \cdot TSR + 10 \cdot P/(1+ODR)$$

LHPR long-horizon progress: grasps and places done / planned

TSR task success · P = (2·LHPR + TSR) / 3

ODR non-target blocks disturbed per counted step (lower is better)



Official demos, ~40 per variant

1 · AGENTS

Understand the task

split it into its skills

approach · close · lift · transport · place · release

2 · AGENTS

Decompose the demos

split demos by skill label

shift segments ±20 mm, ±12°

re-solve joints by IK

→ **16-row action chunks**

3 · TRAINING

Train the small action models

behaviour cloning

skill selector + executor

MLPs, 0.45–17 M params

Oct 13–16

Replica rebuilt from the official HDF5 simulator config: Task 3-2 first, then Tasks 1–3.

Oct 23–28

With the agents: skills, then a program that retargets demos to perceived blocks.

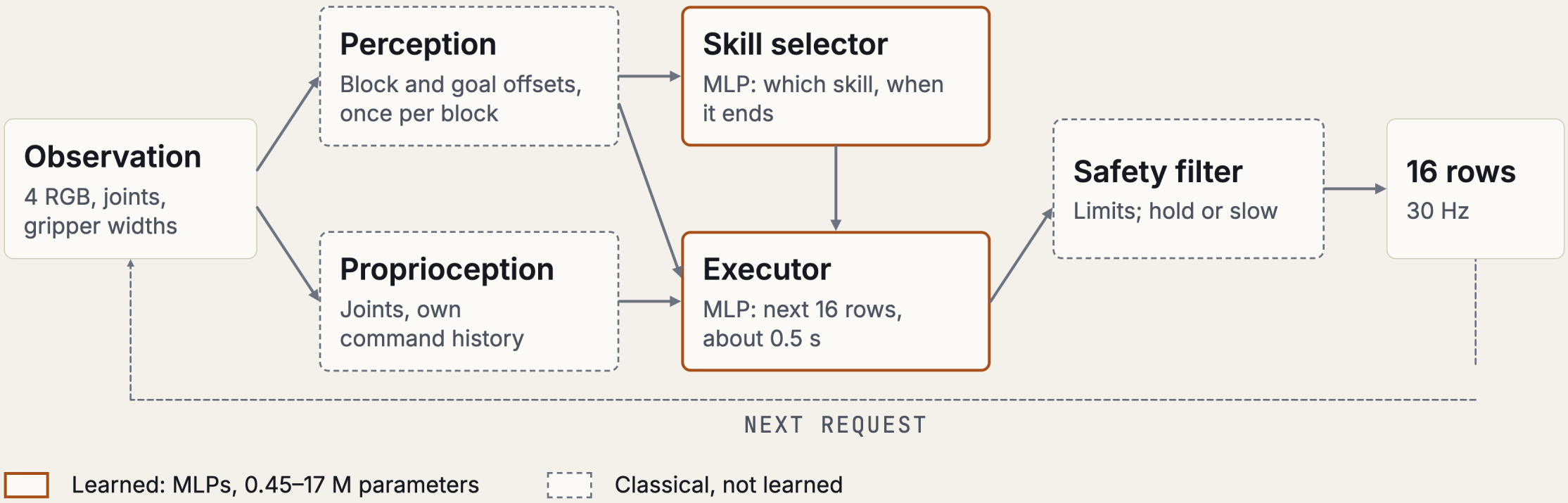
Oct 27

Rule change: learned policies encouraged; no replay, no fixed scripts.

Oct 29 – Nov 1

Decisions and motions moved into learned networks: RAX-AM.

Run time: two small learned modules

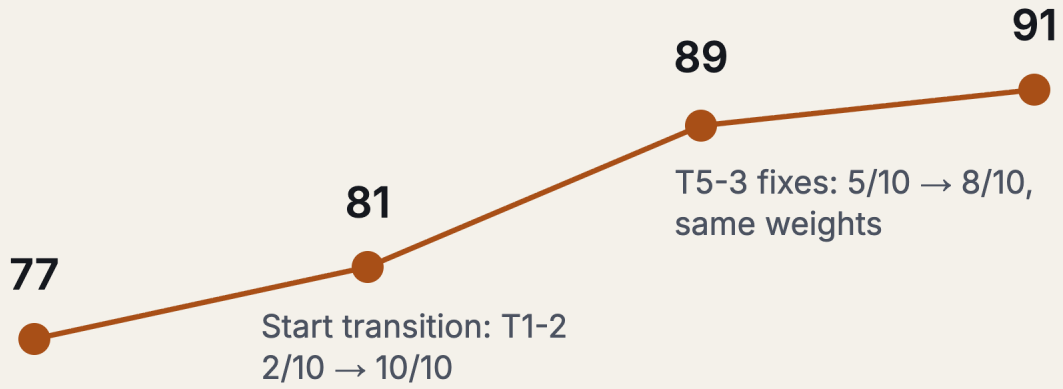


Results: RAX-AM performs best among our submissions

91.93

Best single submission, RAX-AM v6d: 91 of 100 episodes, 6 of 10 variants at 100

Official successful episodes out of 100, by RAX-AM version



98.6 vs 89.4

Tasks 1–3: RAX-AM v6d vs end-to-end π0.5, official

93.50

Per-variant best of our learned submissions (T2-2 from π0.5)

	v1		v3		v6b		v6d
Official	T2-2		T2-3	T3-2	T3-3	T4-2	T4-3
RAX program	91.4		100	100	95.6	80.9	81.3
RAX-AM	91.7		100	100	**100**	**83.91**	**91.9**

Same track and scorer, 10 episodes each · RAX-AM row: v6d, T4-2 from v6b

05 · TAKEAWAYS

Takeaways

1 Human and programmed demonstrations feed the robot agent

Official human demos give the skills and their order. The agents' program retargets them to new block positions, ±20 mm and ±12°.

2 In contact-rich tasks, small learned action models beat the pure agent-supervised program

Task 4 push-down, official, same track: 83.9 and 91.9 vs 80.9 and 81.3.
Task 5 insertion: even.



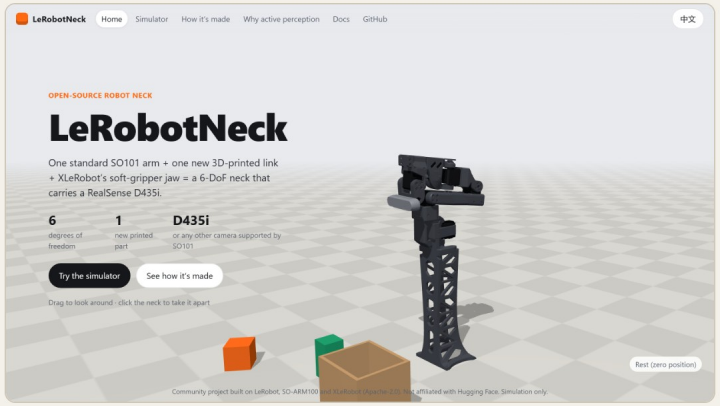
Official eval · RAX-AM v6d · T5-2 · 4×

ABOUT ME

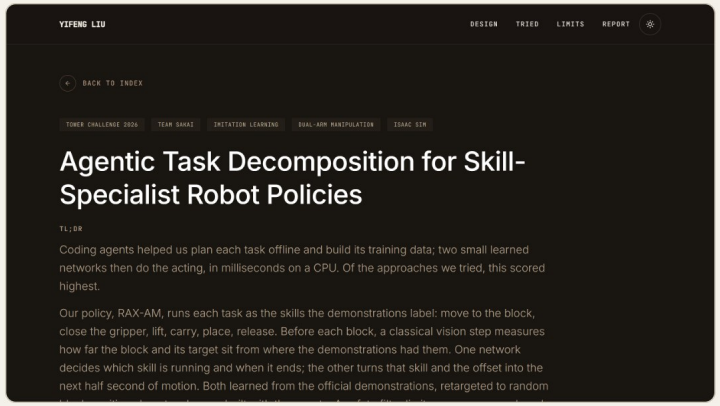
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NOW	Research affiliate, MIT Media Lab, HKUST; Global Traveler
STUDIED	Master's, MIT · Bachelor's, Tsinghua University
RESEARCH	Embodied AI: robot learning, perception and agents
BEFORE	MLE, optimisation and ML at ByteDance and Meituan for recommendation/logistics



LeRobotNeck · open-source robot neck
Hardware and simulation for active perception



TOWER Challenge 2026 · this talk
Agentic robot learning



Act2See · RSS 2026 workshop paper
In submission

Team Sakai · TOWER Challenge 2026



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Yifeng (Sakai) Liu

FULL STACK in robot engineering
from perception to action learning,
from system to hardware

Open to Work!!!

Thanks !